

CHIRAGKUMAR MAKWANA

Robotics Professional with 7 years of industrial robotics experience and MTech in Mechatronics

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SKILLS & ROBOTICS DOMAIN EXPERTISE

Languages - C++, Python

Libraries & Frameworks - PCL, OpenCV, Orocos
KDL, OMPL, Moveit, Behavior Trees, OpenRMF

Operating Systems - Linux, Windows

Simulators - Gazebo

Robotics Middleware - ROS/ROS2, Autoware,
Moveit

Sensors - Camera(Mono, Stereo, depth), IMU, Li-
dars(2d,3d), GPS/GNSS

Drone firmware - Ardupilot, PX4

Cloud - AWS

Hardware Platforms - Nvidia Jetson, Raspberry-Pi,
Intel NUC

Robotics Domain Expertise

UAV 3D navigation

Drone Autonomy using Ardupilot, PX4

Fleet Managements system using OpenRMF and
BehaviorTrees

Sensor Fusion, Mapping

GPS/Non-GPS, Multi-modal navigation, Multi-
source localization

Kinematic Modelling, Gait Planning of legged
robots

EXPERIENCE

Sr. Robotics Engineer- Jio Platform Ltd

July 2025- current

New Mumbai, India

- Implemented real-time visual grounding algorithms to map spatial instructions directly to the humanoid robot's end-effector workspace.
- Boosted inference frame rates by 40% by optimizing detection, segmentation, and 6DoF object pose estimation pipelines leveraging foundation models on Jetson AGX for real-time humanoid manipulation.

Sr. Research Engineer- Novus Hi-Tech Robotic Systemz Pvt. Ltd.

July 2024-July 2025

Gurgaon, India

- Architecting navigation framework for offroad skid steer JCB vehicle with custom path planners, motion controllers and route tracking.
- 2.5D Mapping and Context based multi-gridmaps fusion for offroad vehicle.
- Overseen development of multi information gridmap fusion based on Probabilistic, Evidential and Cv-class based Context methods.

Sr. Robotics Engineer- Acceleration Robotics (TechnoYantra Robotics Pvt Ltd)

January

2021 - June 2024

Pune, India

- Designed, Developed and deployed 2.5D Laser, CV and radar based mapping framework on the car.
- Design of custom local/motion planner for holonomic drive robot.
- Developed custom fleet management system for Agriculture vehicle using RMF ROS2.
- Design of coverage global planner for Agriculture vehicle.
- Developed software modules for scheduling and managing operations of robots using Behavior tree CPP.
- Gazebo simulation environment with interface for scaling up and down fleet of robots dynamically within world.
- Developed autonomy stack for in-campus aerial robots, with sensor-fusion, navigation and landing systems.
- Developed behavioral Landing system and Navigation stack for in-campus delivery drone using PX4, Mavros, BehaviorTrees.CPP
- Developed multi-modal navigation and multi source localization system for outdoor robots in ROS1 and ROS2.

Robotics Engineer- Exadatum Pvt. Ltd.
Pune, India

March 2020 - December 2020

- Developed end to end pipeline for setting up drones on AWS-IoT core using cloud-formation and automated building pre-configured Linux image for Raspberry-pi
- Developed a dockerized operating system based on ROS for mavlink based flight controller(Ardupilot/PX4) with development and release environment features.
- Developed ROS packages for Geo-Fencing, Handling I2C device data and streaming data to cloud via MQTT protocol and AWS kinesis data stream.
- Developed an API wrapped on WebSocket and MAVROS to implement on AWS-lambda for front-end and Drone-OS communication with offline syncing capability
- Designed and 3D modeled the commercial-grade Hexa-copter of 2 kg payload capacity for survey purpose.
- Simulated drone applications in Gazebo with ROS

Consultant (Mechatronics)- Futuring Design Pvt. Ltd.
Pune, India

September 2019 - March 2020

- Research & development of POCs for a sport medical devices.
- Developed geometric and kinematic model to generate and correct the body posture for the exercises
- Developed a hardware and GUI software firmware for sport medical device.
- Wrote python based application to locally store, process and stream data from devices to AWS

Graduate Engineer Trainee- Futuring Design Pvt. Ltd.
Pune, India

August 2018 - August 2019

- Responsible for developing a proof of concept of new products.
- Developed POCs for Sport Medical Device.
- Developed application for capturing data from IMU and load cell, Storing data on SD card and post processing data using python.

EDUCATION

M.Tech. in Mechatronics and Automation- MIT-ADT University, Pune, India 2019- 2021
Relevant Courses- Control Systems, Signal Processing, Modelling and simulation

B.Tech in Mechanical- University of Pune, India 2015 - 2018
Relevant Courses- Engineering Mechanics, Theory of Machines, Mechatronics

RESEARCH PAPERS

Transcutaneous Electrical Nerve Stimulation (TENS) based Assistive Device for Surgeons.
IEEE *April 2021*